FLEET TEAM DIGEST

737MAX-FTD-21-19001

Issue Title : Part Number (P/N) 398908-5/-6 Temperature Control Valve (TCV) / Trim Air Valve (TAV) - No Fault Found (NFF) Removal Investigation

Airplane Model	ATA	Minor Model(s)
737MAX	2160-00	-7, -8, -8200, -9, -BBJ
Other Model(s)	**ECCN**	
737NG	9E991	

Originated Date	Last Revised Date	Created On
02/15/2019	09/03/2026	10/18/2023
Estimated Completion Date	**Next Update Date**	
12/24/2027	12/18/2026	

Revision Description

This FTD article is being revised as follows:

1. Removed outdated/unnecessary information
2. Updates to the Test and Clean Tool effort
3. Updates to the System Integration Lab (SIL) effort
4. Updates to the Honeywell Incoming Test Data Collection effort
5. Updates to the AHMO Service Summary Alert effort

Applicability

All 737 MAX airplanes

Description

This FTD article is intended to track the investigation into the high removal and No Fault Found (NFF) rates of the Temperature Control Valve (TCV)/Trim Air Valve (TAV), Part Number (PN) 398908-5, -6.

Background

Operators report removing TCV/TAVs as a result of the following issues:

1. Hot Cabin Reports/Cabin Temperature Control Issues:
   a. Of the removed TCV/TAVs, a majority are due to item #1 above and are considered to be unnecessary removals. TCV/TAV should not be removed due to Hot Cabin Reports/Cabin Temperature Control Issues unless it is faulted.
2. Left/Right TCV fault during Initiated Built-In Test (IBIT) after pack overheat:
   a. If the pack overheat is not reset (pressing trip reset on P5-10 panel) before IBIT, the controller will fault the associated TCV. This is because during pack overheat, the TCV electrically disconnected from controller via relays.
3. Cabin Rate of Climb (CROC) oscillations:
   a. Considered a valid removal - Reference FIM TASK 21-33-00-810-809 and 737NG-FTD-21-17001 for more information.

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4. Maintenance Light upon landing (Status/Maintenance Message 21-55090/91) due to any of the possible faults:
   a. FC 260 "TCV Open Motor Open Fault"
   b. FC 261 "TCV Close Motor Open Fault"
   c. FC 263 "TCV Close Motor Short Fault"
   d. FC 264 "TCV Seized Fail to Open Fault"
   e. FC 265 "TCV Seized Fail to Close Fault"
   f. FC 266 "TCV Open Limit Switch Fault"

The efforts discussed below will be focusing on Item #4.f above.

Note: For information regarding the history of the PN 398908-x, please reference the 737NG-FTD-21-04002, 737NG-FTD-21-09003, and 737-SL-21-103-B.

Status

Boeing is currently working the TCV/TAV limit switch issue by performing the following tasks:

1. System Integration Lab (SIL) testing of valves to capture FC 266 "TCV Open Limit Switch Fault" occurrences while recording switch voltage/resistance and Integrated Air Systems Controller (IASC) internal data that is not available on-wing. This is to provide additional confidence that FC 266 is occurring due to switch contamination and not a software issue.

2. TCV/TAV Test and Clean box development using Reliable Plane

3. Honeywell in-coming testing using modified version of Reliable Plane noted in #2 above to:
   a. Short Term:
      i. Demonstrate capability to identify switch failures and optimize the cleaning methods/algorithms used for Reliable Plane software development
      ii. Establish maintenance intervals by capturing statistics on contaminated switches such as time on wing prior to detecting an issue, and ability to clean a bad switch
   b. Long Term: Capture switch contamination as a probable cause for removal from aircraft / validate removal reason and reduce No Fault Found (NFF) shop reports.

4. 737 MAX Airplane Health Monitoring Onboard (AHMO) Service Summary Alert development
   a. Creating AHMO Service Summary alerts to track occurrences of simultaneous limit switch
   b. Using the data to determine when simultaneous limit switch is most likely to occur
      i. AHMO data shows most issues occur with the left/right TCV position open limit switch as it rarely gets exercised during service
      ii. AHMO data shows that most occur during low APU bleed condition which provides the opportunity to drive the TCV full open and expose degraded open limit switches - related to open limit switch contamination
      iii. Some occurrences appear to be isolated events that do not repeat - not related to open limit switch contamination
   c. Using the data to determine if trends in switch degradation are easily determined based on age of the valve
   d. Create thresholds to recommend Test and Clean using item #2 above. For non-AHMO and 737NG customers, suggested Test and Clean intervals will be derived from data collected from AHMO reports.

For the most up-to-date detailed information, reference the attached "May-2026 Fleet Team Meeting TCV_TAV Presentation".

Interim Action

Follow the associated Fault Isolation Manual (FIM) tasks.

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Final Action

To be determined pending investigation closure.

Milestones

Note: Target dates are based upon currently available status and have risk of future revision.

1. TCV/TAV Test and Clean Tool On-Wing Validation: Q1 2027 (target)
2. TCV/TAV Test and Clean Tool: Q2 2027 (target)
3. Suggested Cleaning Interval Based on 737 MAX data: Q3 2027 (target)
4. AHMO Service Summary Alert:
   a. Service Summary Alert Development Release: Q2 2027 (target)
   b. Service Summary Alert Threshold: Q3 2027 (target)
   c. Interval Based Test and Clean Threshold: Q3 2027 (target)
5. System Integration Lab (SIL) Data Collection: On-Going Effort
6. Honeywell Incoming Test Data Collection: On-Going Effort

Operator Action

1. Operators can implement Boeing Service Bulletin (SB) 737-21A1203 (R1), and Honeywell Component Service Bulletins (CSB) 398908-21-1638 (R1) and 2740028-21-4566 to fully modify a -5 TCV into the -6 TCV configuration.
   a. This modification eliminates motor connection pins and sockets by directly wiring motor to connector/capacitor (i.e. hardwiring the actuator)
   b. Upgrade to the -6 TCV includes a complete valve overhaul

Supplier Information

Honeywell International Inc.
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U.S.A.
CAGE: 59364
https://www.honeywellaerospace.com/

Reference Categories

Reference Type	Reference Number	Revision Number	Reference Date
Service Bulletin (SB)	Honeywell 398908-21-1638	R1	06/04/2021
Service Letter (SL)	737-SL-21-103	C	
Service Bulletin (SB)	Honeywell 2740028-21-4566		
Fleet Team Digest Article (FTD)	737NG-FTD-21-17001		
Fault Isolation Manual (FIM)	21-33-00-810-809		
Fleet Team Digest Article (FTD)	737NG-FTD-21-04002		
Fleet Team Digest Article (FTD)	737NG-FTD-21-09003		

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Parts List

Part Name	Boeing Part Number	Supplier Name	Supplier Number
Temperature Control Valve		Honeywell	398908-5
Temperature Control Valve		Honeywell	398908-6

Related Categories

- Component Issues : Issues that occur during component maintenance or have an impact on component reliability
- Dispatch Reliability Issues : Issues that cause revenue flight delays and cancellations
- High Occurrence Maintenance Message : null

Attachments:

a) May-2026 Fleet Team Meeting TCV_TAV Presentation

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